25901437 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCPHY202 - BASIC ELECTRONICS AND ELECTRICITY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **10** questions

Each question carries **2** marks

1. Define 1 Volt. [K] / [CO1]
2. Write the expression for instantaneous EMF generated in an AC generator. What will be the EMF when the coil is perpenidular to the field? [U] / [CO1]
3. State Lenz's Law. [U] / [CO1]
4. What is the effect of temperature on electrical resistance of insulators? [U] / [CO1]
5. What is the relation between voltage across resistor and voltage across inductor in an LR circuit? [U] / [CO2]
6. Obtain ripple factor for a half-wave rectifier. [U] / [CO3]
7. Sketch and explain the reverse characteristic of a Zener diode. [U] / [CO3]
8. What are the properties of a PN junction? [U] / [CO3]
9. Give reasons for the permanent destruction of the PN juncion with high reverse voltage. [U] / [CO3]

10. Why is the current gain of a CE configuration greater than unity? [U] / [CO4]

11. What is meant by knee voltage in a common-emitter transistor configuration? [U] / [CO4]

12. Explain how transistor amplifies a weak signal? [A] / [CO4]

13. List the characteristics of an ideal Operational Amplifier. [U] / [CO5]

14. What is an inverting operational amplifier? [U] / [CO5]

[2x10 = 20]

Part B

Short Essay

Answer any **6** questions

Each question carries **5** marks

15. The resistance of a conductor is 12 Ω at 0°C and 14.4 Ω at 50°C. Find the temperature coefficient of resistance of the material. [U] / [CO1]

16. An ac current of frequency 50Hz is applied through a coil containing 200 mH. Calculate the inductive reactance offered to the flow of ac current. [U] / [CO2]

17. Derive the expression for current in an LCR series circuit and discuss its variation with frequency. [U] / [CO2]

18. Explain the mechanism by which a Zener diode keeps the output voltage constant when the input supply voltage varies. [U] / [CO3]

19. A transformer supplies 30 Vrms to a full-wave rectifier. Find the maximum voltage and dc output voltage of a bridge rectifier. [A] / [CO3]

20. Compare the output voltage variation in a rectifier circuit without a filter and with a capacitor filter. [An] / [CO3]

21. Describe with circuit diagram, input and output characteristics of a CE amplifier. [U] / [CO4]

22. Define an operational amplifier and explain its significance in electronic circuits. [U] / [CO5]

[5x6 = 30]